

Kartick Sridhar

sridharkartick29@gmail.com | <https://www.linkedin.com/in/kartick-sridhar-b8b10aa7/>

Irvine, California | 949-880-5932

Career Objective: I am pursuing a career in sensor and chip product development and test engineering, backed by three years of hands-on experience in post-silicon validation. During my graduate studies, I specialized in advanced semiconductor devices, analog and mixed-signal design, and built a strong foundation in digital design. Leveraging my expertise with industry-standard ATE testers such as 93K and Sapphire for wafer and unit testing, along with my proficiency in waveform debugging and JTAG principles, I aim to excel as an analog/mixed-signal test engineer. My collaborative approach and teamwork have consistently driven results and fostered excellence. I am eager to apply these skills to tackle real-world challenges in silicon debug and bridge the gap between simulation and reality.

Work Experience

Product Development Engineer/Post-Silicon Validation
Intel Corporation | Folsom

Feb 2022 - Nov 2024

- My key role involved design verification and validation of SOC, IOE and GPU dies of Intel's latest 14th gen Meteor Lake and Arrow Lake chips using first time chiplet to desktop processors. The role also involved working with various stake holders of other teams across Malaysia, Ireland, Israel, and USA and understanding team dynamics for efficient project and deliverable management.
- End to end silicon engineering from pre-silicon readiness to post silicon bring up and production.
- Schedule capture of layout tools with PCB design experience in all phases of design and testing
- Power management features, mode optimization, voltage margining and checking.
- Understanding of algorithms involved in fixing erroneous bits in memory of CPU chips.
- Python programming for incorporating the logic to enable the overall process of functional boot flow.
- Writing and testing System Level Test cases, creating tester debug tools and infrastructure for inspecting failures
- JTAG methodology for testing various IPs such as PCIE, USB, GBE in different dies.
- Wafer testing and packaging to test multiple dies and units through a thermal socket head and a tester interface unit.
- Analyzing and inspecting test patterns in various stages of reset to improvise yield and reduce test time in wafer sort and class test programs.
- Producing automated test patterns (ATPG) for inspecting manufacturing defects
- UNIX platform for test vector conversion processing and producing test patterns to be loaded on the tester.
- Microsoft Visual Studio for test program flow creation and for building test programs using a GUI.
- GitHub tools and infrastructure to maintain repositories, incorporate changes to test programs from various teams and creating pull requests.
- Reading hardware architecture specification sheets of chip design architecture and collaboration with design and pre silicon testing teams.
- Data analysis tools like Crystal Ball, Microsoft Excel and JMP for analyzing yield data.
- Flash EDA solution for performing full chip validation using STIL (Standard Test Interface Language) and their connectivity.
- Collaborate with design and deployment teams for quality of chips as per design specification.

Software Engineer Trainee

Mar 2021 – Feb 2022

Harvest Software Solutions, LLC | Jacksonville.

- My key role was involved in gathering data, converting project specifications and statements of problems and procedures to detailed logical flow charts to design and develop computer programs for the company's proprietary software products leveraging on techniques of Machine Learning and following the best practices of the software development life cycle (SDLC).
- Building front end client interface using HTML, CSS, Java script, angular JS and Python in back end integration

Data Scientist Intern**Spryte Labs of Byte Consulting Inc. New York City, NY**

Jul 2020 – Mar 2021

- Worked as Data Scientist to wrangle data on cloud and apply Machine Learning models to match clients and tech service providers on the Spryte App used for vendor staff management for large organizations.
- Testing functionality, business requirements and fixing issues in collaboration with developers for new versions roll out

One-year Industrial Experience as part of academics**HCL Technologies (Innovation Labs) Great Britain Ltd, UK**

Jul 2016 -Jun 2017

I was inducted to corporate working during this one year where I learnt about product development and software life cycle end to end. Also, the functional exposure to banking, commodities market and health science helped me learn the ability to adapt technologies to various requirements and hence understand the requirements and its impact on underlying products or its performance.

- Created a Machine Learning model for filtering spam words in the text of official emails using the Naïve Bayes and Gary Robinson's method programmed in JavaScript.
- Worked on an IoT project for motion detection and commodities trading using inertial sensors on a Raspberry-pi for data acquisition and created a front-end web dashboard containing various analytics.
- Supported Deutsche Bank's Transaction Banking applications team interacting with various product experts and Subject Matter Experts.

Summer Intern**HCL Technologies (Innovation Labs) Great Britain Ltd, UK**

Jun 2015 – Sep 2015

- Prepared a scrum board for maintaining a list of various tasks during the software development project pertaining to visitor system management.
- Broke down weekly tasks into various sprints as per the agile methodology.
- Created a friendly front-end user interface using HTML, CSS, and JavaScript for managing visitors.
- Incorporated changes to the UI on a timely basis based on feedback from the administration team.
- Followed UX design strategies in creating the user interface to suit clients of various roles and persona.

Core Skills

Programming languages: C, C++, Java, JavaScript, PHP, Python, MATLAB

PCB Design tools: OrCAD, Altium, NI Multism

Machine learning tools: sklearn, TensorFlow, Pytorch, OpenAI, Mujoco

Operating Systems: Linux, Unix, Windows, ChromeOS, Android Studio

Cloud Technologies: Firebase database, Cloud Firestore, Google Data Studio

Embedded device & sensor programming: PIC microcontrollers, Raspberry-pi, and Arduino

IC Design Tools: Cadence Virtuoso, Keysight Advanced Design System (ADS)

Hardware description languages: VHDL, System Verilog

Version Control tools: Git, Github, Bitbucket

Databases: MySQL, BigQuery

Microsoft tools– MS Word, MS Excel, MS PowerPoint, MS Visual Studio

Silicon Validation tools and techniques: Wafer testing and packaging, ATE testing, JTAG Debugging

EDA: Flash test pattern creation

MATLAB: Analyze data and develop algorithms

Education

University of California, Irvine

Sep 2018 - Dec 2020

Masters | Electrical and Computer Engineering

The University of Manchester, United Kingdom

Sep 2014 - Jul 2018

Bachelors (Honors) | Electrical and Electronic Engineering

Certificates

Cluster Analysis and Unsupervised Learning in Python (Udemy)

Nov 2018

FPGA Design Learning VHDL (Udemy)

Dec 2018

Machine Learning Specialization by Andrew Ng (Coursera)

Jul 2023

VLSI CAD Part II: Layout (Coursera)

Jun 2024

Google AI Essentials (Coursera)

Jan 2025